Leetcode 458. Poor Pigs

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Problem

There are b buckets, one and only one of them is poisonous, while the rest are filled with water. They all look identical. If a pig drinks the poison in one round, it will die at the end of that round and dead pigs cannot participate in subsequent rounds. What is the minimum number of pigs, n^*, you need to figure out which bucket is poisonous within r rounds?

Solution

Let $f(r, n)$ be the maximum number of buckets that can be tested with n pigs in r rounds. We can derive a recursive formula for $f(r, n)$ as follows:

$$f(r, n) = \sum_{k=0}^{n} \binom{n}{n-k} \cdot f(r-1, k)$$

Below, we will show that this closed-form expression for $f(r, n)$ is $(r+1)^n$ and then derive the formula for n^*.

Claim. $\forall r, n \in \mathbb{Z}$, $r, n \geq 0$. $f(r, n) = (r+1)^n$

Proof. We proceed by double induction on r and n, where $r, n \in \mathbb{Z}$ and $r, n \geq 0$, to show $f(r, n) = (r+1)^n$.

Note that the bases cases $f(0, n) = 1$ and $f(r, 0) = 1$ hold for all $r, n \geq 0$ since, with zero pigs or zero rounds, the poisoned bucket can only be identified if it is the only bucket. Similarly, $f(1, 1)$ holds since, with one pig and one round, the poisoned bucket can be identified between $(1+1)^1 = 2$ buckets (but no more).

We will now show that the formula holds for all $r \geq 1$ and $n \geq 1$.

1. $\forall n$. $f(1, n) = (1+1)^n = 2^n$

Assume that $f(1, n) = 2^n$ holds for some $n \geq 1$. We need to show that $f(1, n+1)$ also holds.

Using the recursive definition:

$$f(1, n+1) = \sum_{k=0}^{n+1} \binom{n+1}{n+1-k} \cdot f(0, k)$$

Since $f(0, k) = 1$ for all k, we have:

$$f(1, n+1) = \sum_{k=0}^{n+1} \binom{n+1}{k} = 2^{n+1}$$

The right-hand side is:

$$(1+1)^{n+1} = 2^{n+1}$$

Thus, the statement holds for $n+1$.

2. $\forall r.\ f(r, 1) = (r+1)^1 = r+1$

Assume that $f(r, 1) = (r+1)$ holds for some $r \geq 1$. We need to show that $f(r+1, 1)$ also holds.

Using the recursive definition:

$$f(r+1, 1) = \sum_{k=0}^{1} \binom{1}{1-k} \cdot f(r, k)$$

Expanding this gives:

$$f(r+1, 1) = \binom{1}{1} \cdot f(r, 0) + \binom{1}{0} \cdot f(r, 1)$$

Since $f(r, 0) = 1$ and by the inductive hypothesis $f(r, 1) = r+1$, we have:

$$f(r+1, 1) = 1 \cdot 1 + 1 \cdot (r+1) = r+2$$

The right-hand side is:

$$(r+2)^1 = r+2$$

Thus, the statement holds for $r+1$.

3. $\forall r, n.\ f(r, n) = (r+1)^n$

Assume that $f(r-1, n) = r^n$ and $f(r, n-1) = (r+1)^{n-1}$ hold for some $r, n \geq 1$. We need to show that $f(r, n) = (r+1)^n$.

Using the recursive definition:

$$f(r, n) = \sum_{k=0}^{n} \binom{n}{n-k} \cdot f(r-1, k)$$

By the assumption $f(r-1,k) = r^k$, we get:

$$f(r,n) = \sum_{k=0}^{n} \binom{n}{k} \cdot r^k$$

This is the binomial expansion of $(r+1)^n$, so:

$$f(r,n) = (r+1)^n$$

By the principle of double induction, the formula $f(r,n) = (r+1)^n$ holds for all $r \geq 0$ and $n \geq 0$.

$\square$

n^* is the smallest integer n such that $b \leq (r+1)^n$. Rearranging the inequality, we get $n \geq \lceil \log_{r+1} b \rceil$. Therefore, $\boxed{n^* = \lceil \log_{r+1} b \rceil}$.